

Question ID: f5ff91b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question
If $\frac{x-5}{7} = \frac{x-5}{9}$, the value of $x - 5$ is between which of the following pairs of values?

- Answer**
- A. -9 and -7
- B. -3 and 3
- C. 4.5 and 5.5
- D. 6.75 and 9.25

Correct Answer: B

Rationale
Choice B is correct. Multiplying both sides of the given equation by $(7)(9)$, or **63**, yields $(63)(\frac{x-5}{7}) = (63)(\frac{x-5}{9})$, or $9(x - 5) = 7(x - 5)$. Subtracting $7(x - 5)$ from both sides of this equation yields $2(x - 5) = 0$. Dividing both sides of this equation by **2** yields $x - 5 = 0$. Therefore, if $\frac{x-5}{7} = \frac{x-5}{9}$, then the value of $x - 5$ is **0**. It follows that of the given choices, the value of $x - 5$ is between -3 and **3**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.